25900685 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCZGY203 - BIOLOGICAL BASIS OF BEHAVIOR - III

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Which of the following best describes homeostasis? [K] / [CO1]

   a). The ability of an organism to change permanently

   b). The process of maintaining internal balance despite external changes

   c). The conversion of nutrients into energy

   d). The growth and reproduction of organisms

2. The nerve that carries smell sensation to the brain is: [K] / [CO1, CO2]

   a). Optic Nerve

   b). Olfactory Nerve

   c). Auditory nerve

   d). Facial nerve

3. Bowman’s glands in the olfactory epithelium secrete: [U] / [CO1, CO2]

a). Neurotransmitters

b). Mucus and odorant-binding proteins

c). Enzymes for action potential generation

d). Growth hormones

4. Which of the following statements is TRUE regarding Charles Bonnet Syndrome? [U] / [CO3]

a). It indicates underlying dementia.

b). It requires antipsychotic treatment in all cases.

c). It occurs only in children.

d). Patients usually maintain normal cognitive function.

5. Which part of the brain processes temperature sensations? [U] / [CO3]

a). Photoreceptors activate

b). Mechano receptors activate

c). Somatosensory cortex

d). Chemoreceptors activate

6. Hypermetropia can be corrected by using: [U] / [CO3]

a). Concave lens

b). Convex lens

c). Cylindrical lens

d). Contact lens only

7. Which of the following hormones has a direct effect on blood pressure? [U] / [CO4]

a). Oxytocin b). Vasopressin c). Prolactin d). Thyroxine

8. Dysregulation of the HPA axis can lead to: [U] / [CO4]

a). Enhanced stress resilience

b). Hyperthyroidism

c). Hypothyroidism

d). Reduced stress management

9. Postpartum depression is classified under: [U] / [CO4]

a). Anxiety disorders

b). Mood disorders

c). Psychotic disorders

d). Neurodevelopmental disorders

10. Dysregulation of which axis is commonly associated with anxiety disorders? [U] / [CO4]

a). HPG axis b). HPT axis

c). HPA axis d). GH axis

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. State the cranial nerves involved in carrying taste signals. [U] / [CO2]

12. Enumerate the functions of taste buds. [U] / [CO1, CO2]

13. State the function of rod cells. [U] / [CO3]

14. Which is the layer of epidermis present in thick skin and comment on it? [U] / [CO3]

15. Give the pathway of visual impulses from the retina to the visual cortex. [U] / [CO3]

16. Describe the function of the middle ear. [U] / [CO1, CO2]

17. Mention the ordered sequence of auditory relay stations from periphery to cortex. [U] / [CO2]

18. A woman with PCOD complains of infertility. How do assisted reproductive techniques apply in her management? [A] / [CO4]

19. Mention two common physical symptoms of premenstrual dysphoric disorder. [K] / [CO4]

20. Why is the pituitary gland called the “master gland”? [K] / [CO4]

21. Write short notes on Graves’ disease. [U] / [CO4]

22. Name one mood stabilizer used in the treatment of bipolar disorder. [K] / [CO4]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. Describe the functional importance of olfaction in daily life activities. [U] / [CO1]

24. Explain the different primary taste sensations and give examples. [U] / [CO2]

25. Explain how nociceptors are involved in the perception of pain through the skin. [U] / [CO3]

26. Compare Meissner's and Pacinian corpuscles? [U] / [CO3]

27. Describe the important functions of endocrine system in our body. [U] / [CO4]

28. Explain the interaction between sympathetic nervous system and adrenal medulla in stress. [U] / [CO4]

29. Discuss how hormonal changes in the ovaries prepare the female body for fertilization and implantation. [U] / [CO4]

[4x5 = 20]